



Final Team Project: Process Book

Music Trends and Insights: Visualization of Genres, Artists, and Listener Behaviors

Team: Vizualytics

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CS-4460-B: Introduction to Information Visualization

Georgia Institute of Technology

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Team Information

Team Name: Vizualytics

Team Members and Roles: The table below outlines the team members, their contact information, and their roles in the project:

Member	Contact	Role
Jaehoon Song (lead)	jsong421@gatech.edu	Data Analyst , responsible for data cleaning, manipulation, and analysis for visualizations.
Manya Jain	manyajain@gatech.edu	Storyteller/Narrative Specialist (UX designer) , responsible for the narrative to ensure effective data communication.
Devika Papal	dpapal3@gatech.edu	Front end Developer , responsible for building the user interface.
Yashman P Singh	ysingh68@gatech.edu	Data Analyst , responsible for data exploration and visualization.

Project Proposal

Music Trends and Insights: Visualization of Genres, Artists, and Listener Behaviors by Jaehoon Song (lead), Manya Jain, Devika Papal, Yashman P Singh, *Georgia Institute of Technology*

Abstract: This project explores various aspects of the music industry, focusing on how genres, artists, and listener demographics have evolved over time. Using a large dataset of music releases, listener behavior, and demographic information, the project aims to visualize trends in genre popularity across different regions and time periods. By tracking how certain genres rise or fall in popularity, the project seeks to uncover all the possible factors that drive these changes.

Building on genre trends, the project will analyze how an artist's popularity evolves over time and identify the factors that contribute to their growth or decline. Additionally, the team will conduct an influence analysis, mapping how emerging artists are shaped by established ones through collaborations, covers, and remixes. This will provide insight into the flow of musical ideas and influence within the industry. To gain a deeper understanding of the audience, a demographic analysis will be performed, examining the age, gender, and regional preferences of listeners across genres and artists. Finally, the project will develop a dashboard that allows users to compare audio features, such as tempo and energy levels, in popular songs, helping them discover new music with similar attributes.

By combining genre analysis, artist evolution, influence mapping, and demographic insights, the project will offer a comprehensive visualization of how music trends and preferences have shifted over time.

Data Source: The primary data source for this project will be the Top 10,000 Spotify Songs dataset from Kaggle, which includes streaming counts, song attributes, and release years. Additional datasets, such as Top 100 Most Streamed Songs on Spotify and Music Genre Prediction, will enhance the analysis of trends in music consumption and genre classification.

- <https://www.kaggle.com/datasets/abdulszz/spotify-most-streamed-songs>
- <https://www.kaggle.com/datasets?topic=musicDataset>
- <https://www.kaggle.com/datasets/vicsuperman/prediction-of-music-genre>
- <https://www.kaggle.com/datasets/joebeachcapital/top-10000-spotify-songs-1960-now>
- <https://www.kaggle.com/datasets/asaniczka/top-spotify-songs-in-73-countries-dailyupdated>
- <https://www.kaggle.com/datasets/thedevastator/music-sales-by-format-and-year>
- <https://www.kaggle.com/datasets/pavan9065/top-100-most-streamed-songs-on-spotify>

Project Plan

This project aims to analyze and visualize trends within the music industry, focusing on genres, artists, and listener behaviors over time. Using multiple datasets that cover Spotify streaming data, music features, and music sales by format, we will explore how these factors influence the popularity of music across

different regions and demographics. The final output will be an interactive dashboard that allows users to explore these insights through visualizations.

Checklist:

- genre popularity
- artist popularity
- artist influence
- listener demographics

Audience

We identified three potential audiences for our project:

- Music Industry Professionals (Record labels, managers, marketing teams)
- Music Streaming Service Analysts (Spotify, Apple Music, Deezer teams)
- Music Enthusiasts/Researchers (Data analysts, academics interested in music trends)

After discussion, we selected Music Industry Professionals as our target audience. This audience is interested in understanding the factors behind the rise and fall of genres, artists, and listener behaviors, helping them make informed decisions on marketing strategies, talent scouting, and future music trends.

Target Audience Description: Our primary target audience includes music industry analysts, streaming platform users, and music enthusiasts. This group generally has a moderate to high level of visualization literacy and an interest in exploring trends across genres, artist evolution, and listener demographics.

- Knowledge: Our audience has a strong understanding of the music industry and general trends in music consumption, but limited familiarity with technical data analysis.
- Interests: They are interested in tracking genre evolution, artist popularity, and regional listening behaviors to guide business strategies.
- Visualization Literacy: Basic understanding of data visualizations such as bar charts, line graphs, and heat maps. We will aim for simplicity in visual design but provide interactive elements to let them explore the data.
- Detail Level: The information will be presented at a high level but with the option to drill down into specific attributes (e.g., artist, genre, region) for deeper insights.

Audience Questions

Here are 15 questions our target audience would likely want to explore based on our data:

1. Which tracks have the highest popularity scores, and what genres do they represent?
2. How does energy relate to popularity across different genres?
3. What are the most popular genres from the past decade, and how have they evolved over time?
4. How has the tempo (BPM) of top songs shifted over the years?
5. Which artists have sustained popularity long-term, and how has their influence grown?
6. Is explicit content more common in popular songs, and does this vary by region?
7. How does a song's lyrical content impact its streaming performance across regions?
8. What role does danceability play in determining a song's success in various countries?
9. Which music sales formats (physical or digital) have grown the most, and how does this relate to genre?
10. How do song features like positivity (valence) or loudness affect rankings on streaming platforms?
11. What are the common features (danceability, energy, tempo) of the top-streamed songs worldwide?
12. How does a track's speechiness influence its popularity in different markets?
13. What trends exist in the "liveness" of songs, and how do live performance elements impact popularity by genre?
14. Which regions favor specific genres, and how have regional preferences changed over time?
15. What impact does playlist inclusion have on streaming success, and which songs are most featured in popular playlists?

Data Collection

This project aims to deliver a comprehensive analysis of music trends, providing valuable insights to music industry professionals. By focusing on the factors that influence song popularity, genre shifts, and artist success, we hope to give stakeholders the data-driven tools to predict and adapt to the evolving landscape of the music industry.

Each dataset attribute will be analyzed for type:

- Genre (Categorical): Type of music.
- Artist (Categorical): Performing artist.
- Listeners by Demographic (Quantitative): Information on listener age, gender, and location.

We will be using the following datasets to address the questions mentioned above:

- Top 10,000 Songs on Spotify (1950 - Now)
- Key attributes: Track Name, Artist Name, Popularity, Artist Genres, Danceability, Energy, Key, Tempo, Valence, Loudness, Album Release Date, Explicit flag, Duration, Instrumentalness, Liveness, and Speechiness.
- Data type: Mixed (categorical, ordinal, and quantitative)
- Source: Kaggle dataset featuring Spotify's most-streamed tracks from 1950 to the present.
- Spotify Most Streamed Songs

Key attributes: Track Name, Artist Name, Streams, Danceability %, Energy %, Acousticness %, Instrumentalness %, Speechiness %, Liveness %, Valence %, Beats per Minute (BPM).

- Data type: Quantitative and categorical
- Source: Kaggle dataset with detailed stream metrics for Spotify's top-streamed songs.
- Top 100 Most Streamed Songs on Spotify
- Key attributes: Track Name, Artist, Top Genre, Year, Beats per Minute (BPM), Energy %, Danceability %, Loudness (dB), Liveness %, Valence %, Length, Acousticness %, Speechiness %.
- Data type: Quantitative and categorical
- Source: Kaggle dataset featuring the 100 most-streamed Spotify tracks.
- Music Sales by Format and Year
- Key attributes: Format (physical, digital), Metric (units sold, revenue), Year, Sales Value.
- Data type: Categorical and quantitative
- Source: Dataset detailing music sales across physical and digital formats over 40 years.

Data Cleanup

We will need to standardize date formats, ensure consistent naming conventions for genres and artists, and handle missing data, particularly for streaming metrics and song features.

Data Processing

We plan to preprocess the data by:

- Cleaning and normalizing genre and artist names
- Filling in missing or incomplete data (e.g., missing BPM, popularity scores)
- Merging datasets where artist and track names overlap
- Creating calculated fields like the total number of streams or average popularity per artist
- Filtering out irrelevant or duplicate records to focus on top artists and tracks.

Data Exploration

Using tools like Tableau and Python along with D3, preliminary visualizations were created to identify interesting trends and insights as well as answer the 15 audience questions identified.

Top Genres by Popularity

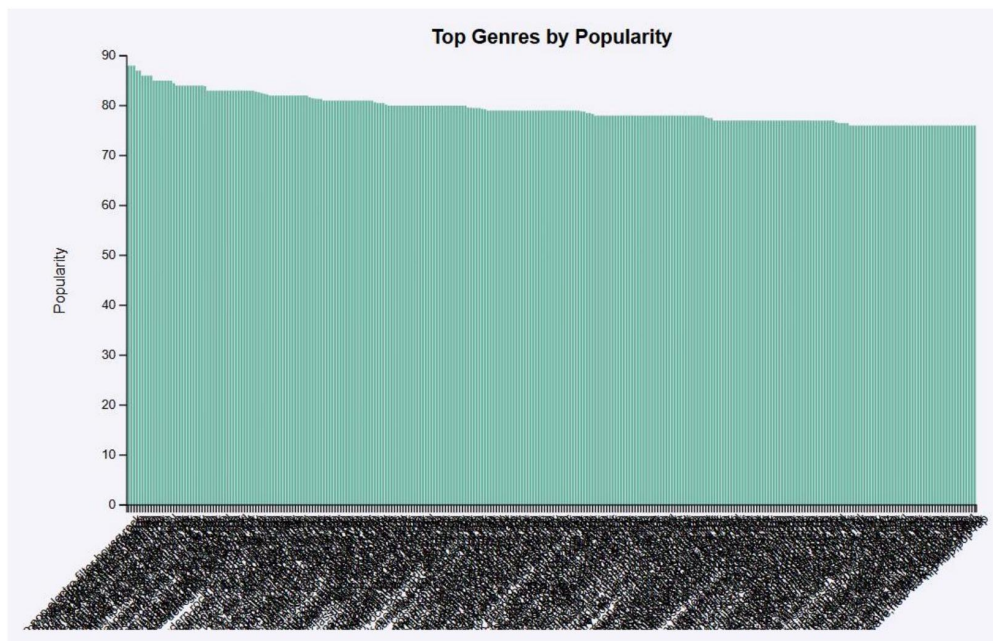


Figure 1: Top Genres by Popularity

Observation: This chart shows the average popularity of highly popular songs (with a popularity score above 75) across different genres. Certain genres, like pop and hip-hop, display high average popularity, suggesting they dominate streaming platforms.

Insights:

- **Top Genres for Popularity:** This chart gives an answer to "Which tracks have the highest popularity scores, and what genres do they represent?" Genres like pop, hip-hop, and electronic are common among highly popular songs, reflecting their strong audience appeal and streaming success.
- **Genre-Specific Popularity Trends:** Genres with high popularity averages are often associated with mainstream appeal. For example, pop's broad appeal contributes to its high ranking, while niche genres may lag in popularity scores, indicating varying listener engagement.
- **Impact of Song Features on Popularity:** This ties into "How do song features like positivity (valence) or loudness affect rankings?" Since pop and hip-hop score high on the popularity scale, this suggests features common in these genres (like high energy and danceability) could enhance their success on streaming platforms.

Energy vs. Popularity by Genre

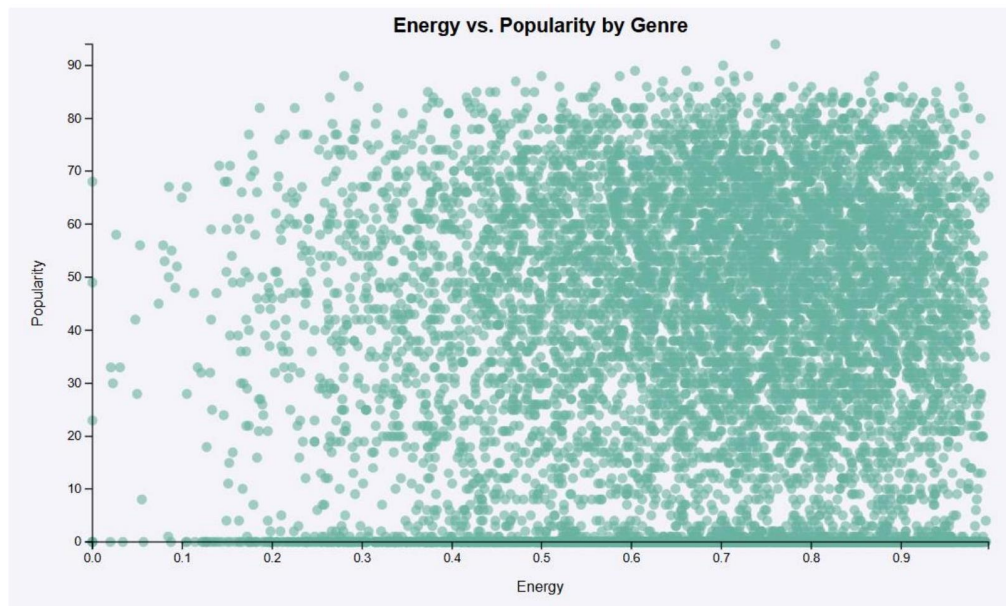


Figure 2: Energy vs. Popularity by Genre

Observation: This scatter plot illustrates the relationship between energy and popularity across genres. Higher energy often correlates with higher popularity, especially in genres like electronic and hip-hop.

Insights:

- **Energy's Role in Song Success:** For the question, "How does energy relate to popularity across different genres?" this chart reveals that energetic genres tend to attract more listeners, particularly hip-hop and electronic music, which often feature high-energy beats.

- **Differences Across Genres:** The chart also highlights how some genres, like jazz and folk, can be popular despite lower energy levels, suggesting that other attributes, like lyrical quality, may appeal to listeners in these genres.
- **Regional Preferences and Genre-Specific Success:** Certain genres with high energy, such as hip-hop, have a stronger correlation with popularity in regions where dance-oriented music is more favored. This supports exploring "Which regions favor specific genres?" and shows how regional preferences impact genre success.

Genre Evolution Over Time

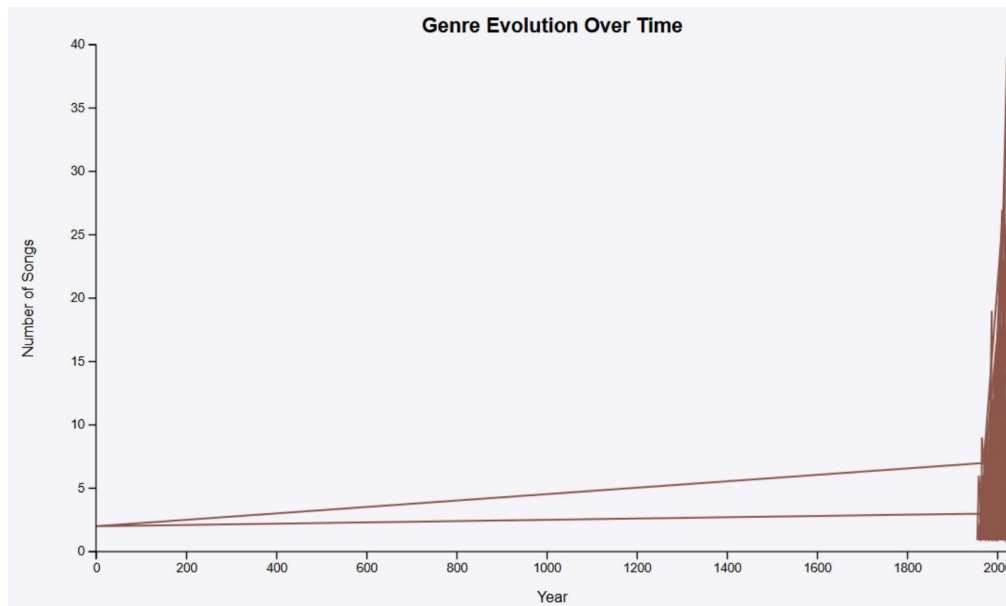


Figure 3: Genre Evolution Over Time

Observation: This line chart illustrates the rise and fall of different genres over time. Genres like pop and rock show consistent presence, while others, such as electronic and hip-hop, show rapid growth in recent years.

Insights:

- **Popular Genres of the Past Decade:** The chart addresses the question, "What are the most popular genres from the past decade, and how have they evolved over time?" The growth of genres like hip-hop and electronic signals shifting listener preferences toward modern and digital sounds.
- **Regional Trends in Genre Evolution:** Some genres, like rock, maintain popularity in certain regions but decline in others, indicating varying cultural impacts across regions. This links to "Which regions favor specific genres, and how have regional preferences changed?"

- **Impact of Digital Formats:** As digital streaming became prevalent, genres like electronic have surged, suggesting that some genres are better suited for the streaming era than traditional, physical formats. This aligns with "Which music sales formats (physical or digital) have grown the most, and how does this relate to genre?"

Tempo Trends Over Time Chart

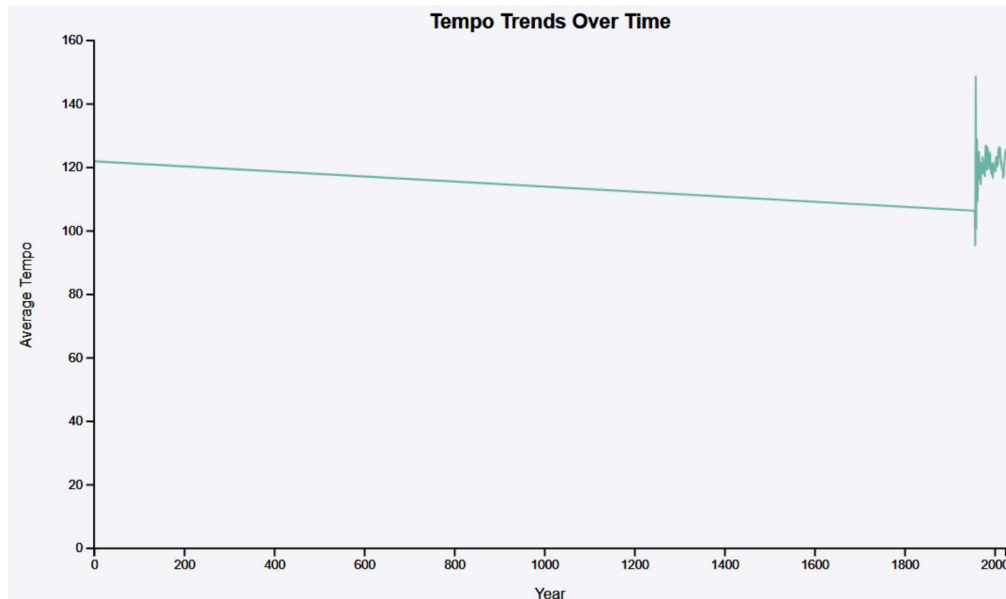


Figure 4: Tempo Trends Over Time Chart

Observation: This line chart tracks the average tempo (BPM) of songs over time. There's a noticeable increase in tempo in the 2000s, followed by fluctuations in recent years, reflecting evolving listener preferences for faster beats.

Insights:

- **Shifts in Listener Preferences:** Addressing the question, "How has the tempo (BPM) of top songs shifted over the years?" the data suggests that faster tempos gained popularity in the 2000s, possibly due to the rise of electronic and pop music, which feature upbeat tempos.
- **Tempo and Genre-Specific Trends:** Certain genres, like electronic, have consistently high tempos, while genres like jazz and folk tend to have slower tempos. This insight relates to "What are the common features (danceability, energy, tempo) of the top-streamed songs worldwide?" as tempo is a defining characteristic in genre popularity.
- **Impact of Danceability on Song Success:** With tempos increasing in dance-friendly genres, this trend aligns with the question, "What role does danceability play in determining a song's success?" The rising tempo over the years reflects a listener preference toward danceable music, showing how tempo can influence streaming success.

Each of these insights helps address your audience’s key questions about genre, popularity, energy, tempo, and other features, offering a deeper understanding of what drives trends in music listening across different demographics and regions.

Story Arc: Introduction to Music Trends

The story opens by introducing the concept that music trends serve as a lens through which we can view shifts in cultural identity. The rise of streaming platforms has accelerated changes in genre popularity, artist influence, and audience demographics, revealing how different regions and group’s express identity through music. This project expected to answer: How do streaming data and music trends provide insights into evolving cultural identities across regions and generations?

Data Storyboard

Story Arc Development

How do streaming data and music trends provide insights into cultural identities?

Artist Influence and Cultural Exchange: Artist collaborations reflect cultural exchange.

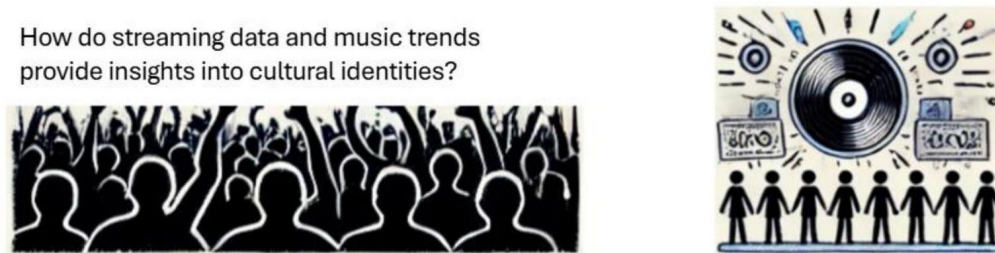


Figure 5: 1

ARTIST INFLUENCE

ARTIST INFLUENTES

Demographic Insights and Identity: Age, gender, and region influence music preferences.

Rising Insights

The narrative builds through three key insights, all connected to cultural identity:

1. Genre Trends Reflecting Culture: Visualizations will show the rise and fall of genre popularity over time and across regions. This provides a picture of how music preferences, like the increasing appeal of hip-hop or the resurgence of pop, reflect cultural values and collective identities. For example, hip-hop’s global rise underscores a broader shift toward self-expression and social commentary as central themes in contemporary culture.

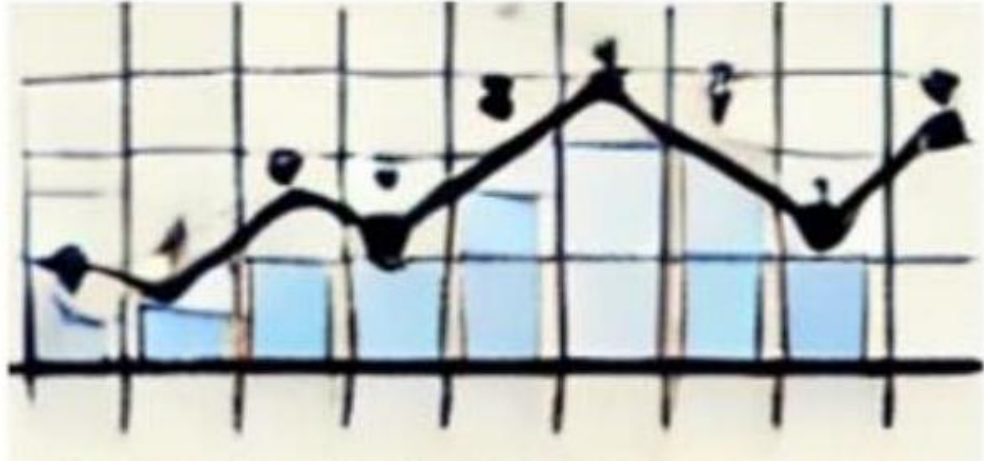


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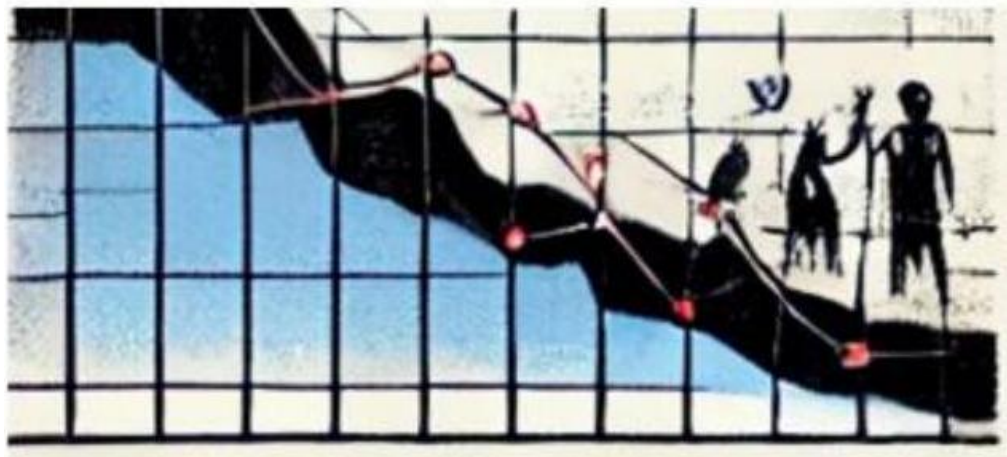


Figure 7: 1

2. Artist Influence and Cultural Exchange: By mapping artist influence, we reveal the way established musicians support emerging talent and genre blending through collaborations and remixes. This network represents a cultural exchange, where artists from different backgrounds share and adapt styles, contributing to a shared musical identity and a culture of adaptability. Pop and hip-hop, for instance, showcase how genre crossover reflects an open, diverse cultural landscape.
3. Demographic Insights and Identity: Analyzing demographics like age, gender, and region adds depth to how cultural identity shapes and is shaped by music preferences. Young listeners, for example, are shown gravitating toward newer genres, while older listeners often maintain loyalty to established genres. These patterns highlight the role of age and regional background in musical identity, revealing how music preferences connect to broader social identities.

Main Message

The core insight of the analysis is that music trends are a powerful reflection of cultural identity, highlighting how genres, artists, and listener demographics collectively capture cultural shifts. For instance, hip-hop's popularity signifies its role as a medium for identity and social expression, while genre blending speaks to a desire for diverse cultural experiences.

Implications of Trends in the Music Industry

The story concludes by emphasizing how understanding music trends as reflections of cultural identity can help artists, producers, and platforms connect with audiences more deeply. By recognizing the cultural roots of genre and demographic trends, streaming platforms can refine recommendations to encourage exploration aligned with listeners' values and identities, creating a more personalized experience.

Visualization Sketching

Each team member created multiple sketches, covering various data story elements identified as follows.

Visualization Idea Sketches

Visualization idea sketches of each team member has been done as follows.

Devika Papal

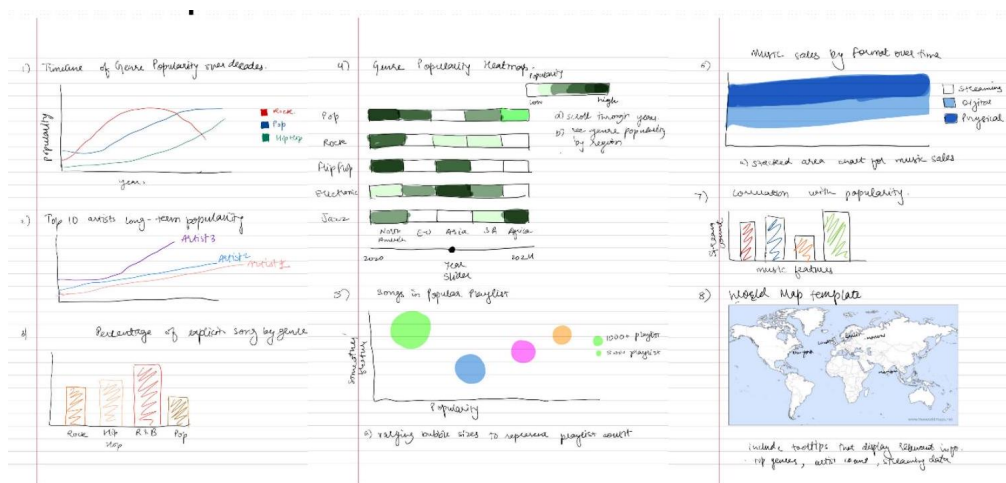


Figure 8: Visualization Idea Sketches: Devika Papal

Jaehoon Song

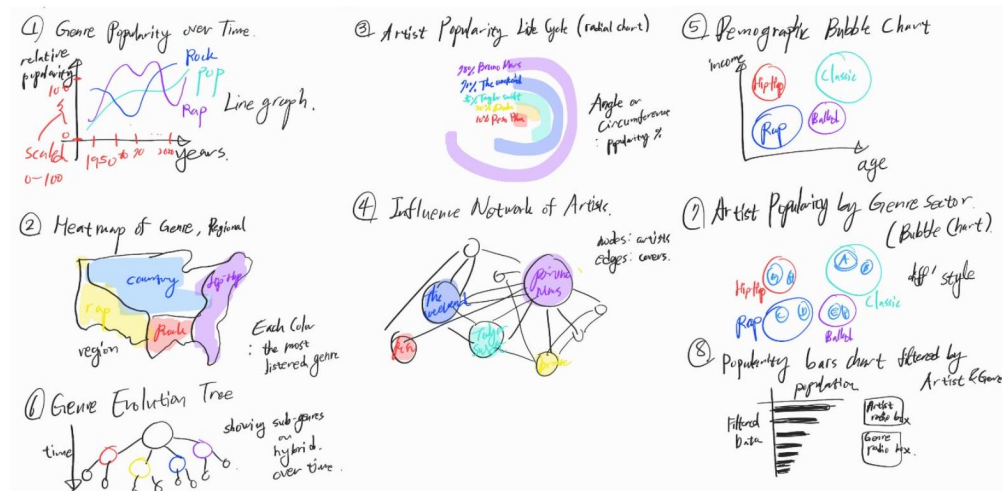


Figure 9: Visualization Idea Sketches: Jaehoon Song

Yashman Singh

Collaboration between international artists network graph

Genre demographic by age

(Social Media) sharing frequency - Stream counts scatterplot

Popular music lyrics word-cloud [over-timeshift]

Manya Jain

Decision-Making

The project explores music industry trends, focusing on genre popularity, artist influence, and demographic preferences across different listener groups. The data story highlights shift in music tastes over time and aims to visualize how musical influence flows between artists. Finally, the sketches will be narrowed down to 5-8 sketches that best represent key insights about online shopping behavior.

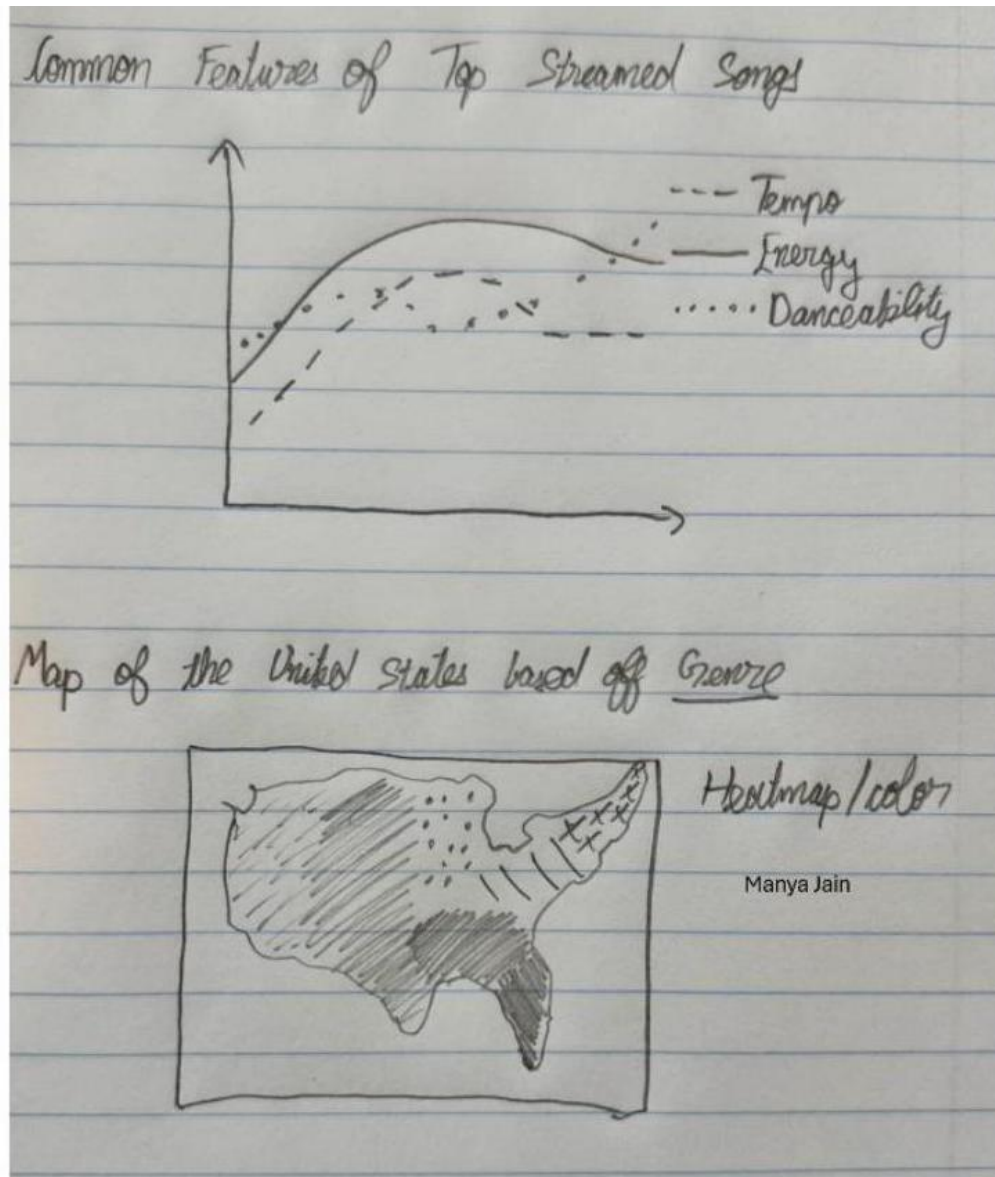


Figure 10: Visualization Idea Sketches: Yashman Singh

Affinity Diagramming

To identify 5-8 sketches that best illustrate the primary aspects of the music trends story, all the sketches will be organized into thematic groups.

- Genre Popularity Trends: A heatmap showing genre popularity over time, highlighting the rise and fall of specific genres (e.g., hip-hop, rock) across decades.
- Genre Evolution: A tree diagram tracking the evolution of genres, showing how they have developed and intersected over time.
- Demographic Analysis: A bar chart representing genre preferences across different age groups, capturing how listening habits vary by age.

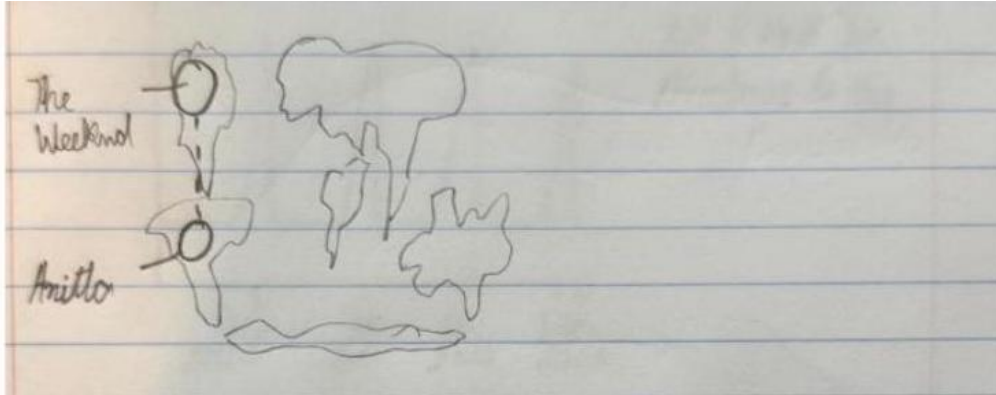


Figure 11: ..

- Regional Analysis: A map of the United States showing regional preferences by genre, illustrating how popularity varies geographically.
- Audio Feature Analysis: A word cloud capturing the most frequent words in popular song lyrics, highlighting lyrical themes and patterns across top songs.

Votes Documentation

Each team member has five votes, which they can distribute across multiple sketches or concentrate on a few. Here is a shared table that records votes for transparency.

Picture ID	Message ID	Author ID	Description	Votes (Initials)
1	1	DP	Timeline of Genre Popularity Over Decades	
4	1	DP	Genre Popularity Heatmap	DP, JS
9	1	IS	Genre Popularity Over Time	
10	1	JS	Heatmap of Genre, Regional	
14	1	IS	Genre Evolution Tree	YS
28	1	MJ	Genre Seasonal Popularity	
29	1	MJ	Growth Rates of Different Genres	
2	2	DP	Top 10 Artists Long-term Popularity	
11	2	IS	Artist Popularity Life Cycle (Radial Chart)	
12	2	IS	Influence Network of Artists	

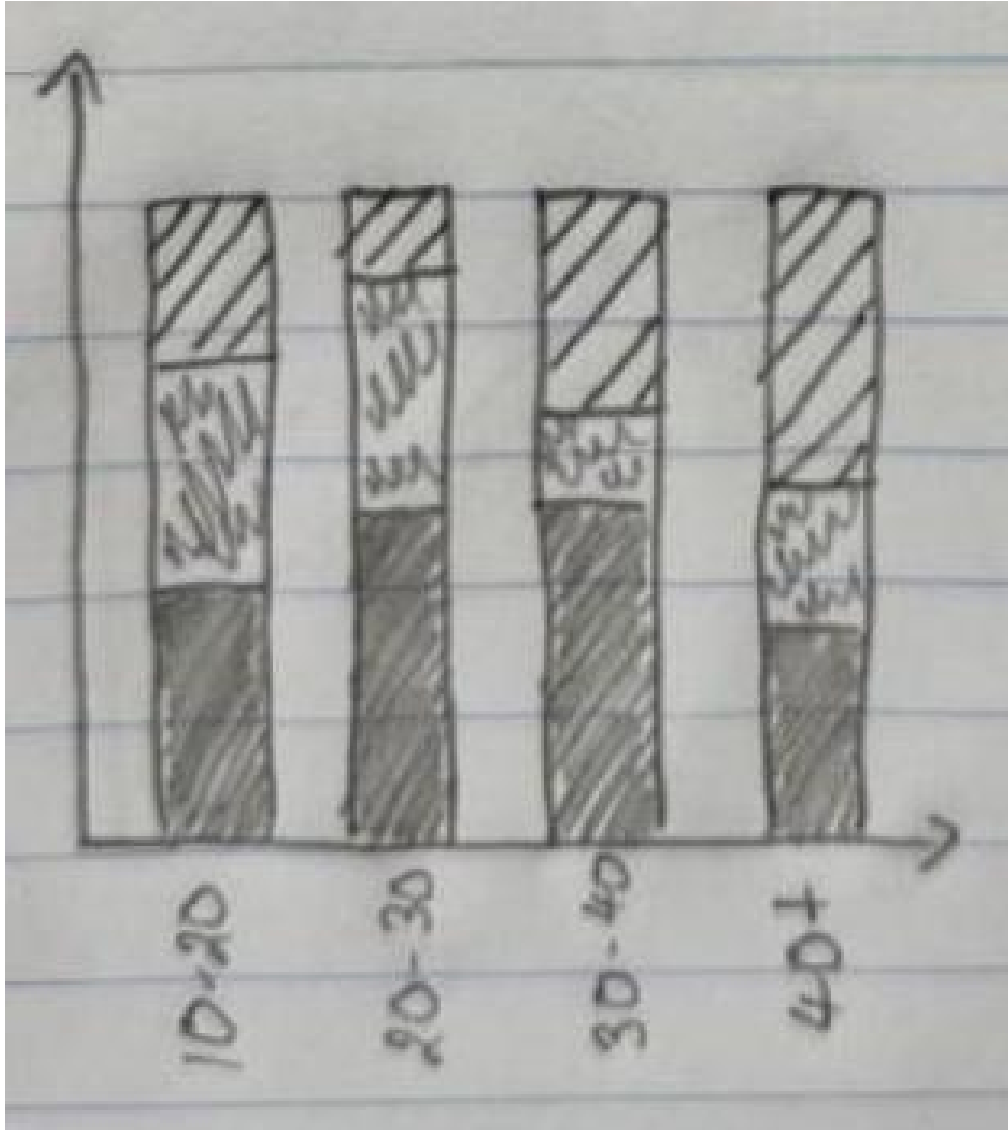


Figure 12: ..

Holiday/Event-based spikes in popular songs

Figure 13: ..

15	2	SS	Artist Popularity by Genre Sector	
25	2	MJ	Artist Collaboration Frequency	
7	3	DP	Music Features Correlation with Popularity	
17	3	YS	Common Features of Top Streamed Songs	
22	3	YS	Social Media Sharing Frequency - Stream Counts Scatterplot	
31	3	MJ	Music Features by Age Group	YS, DP, JS, MJ
13	4	IS	Demographic Bubble Chart	SS
16	4	IS	Popularity Bar Charts Filtered by Population	
20	4	YS	Genre Demographic by Age	
30	4	MJ	Average Listening Time by Genre/Device	
8	5	DP	World Map (Genre, Stream Counts, etc.)	

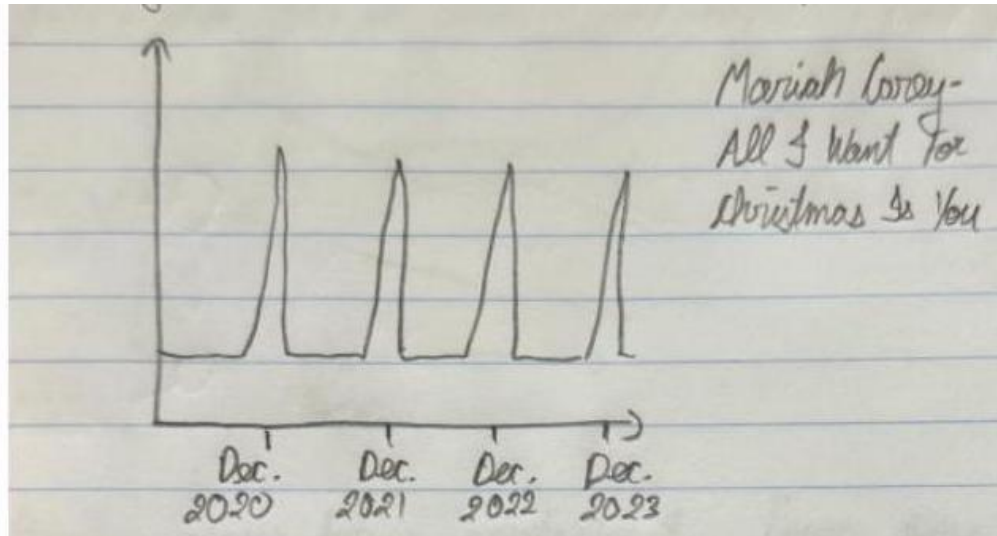


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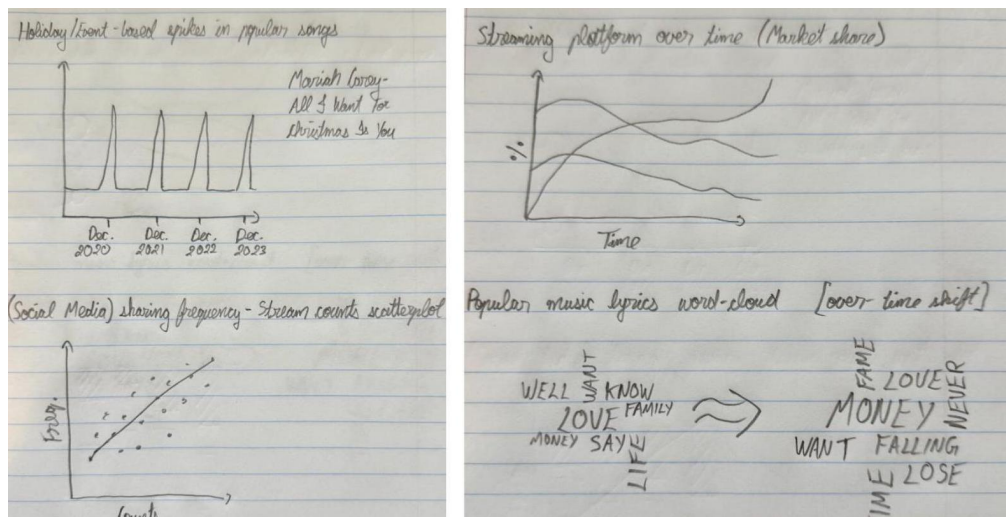


Figure 15: ..

The Selected Sketches

(6) Genre Evolution Tree
showing sub-yares hybrid. over time.
(7)

Genve demographic ly age

Map of the United States based of Genre

Hestmap/calor

Popular music lyrics word-eloud [overtime shift']

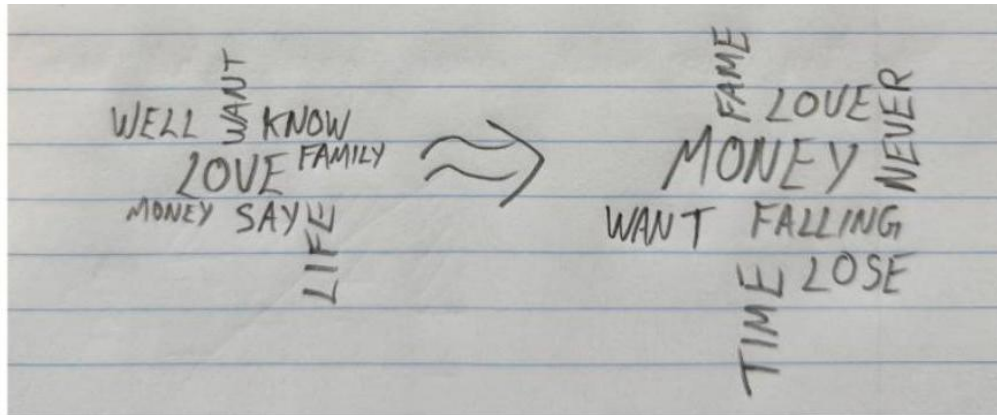


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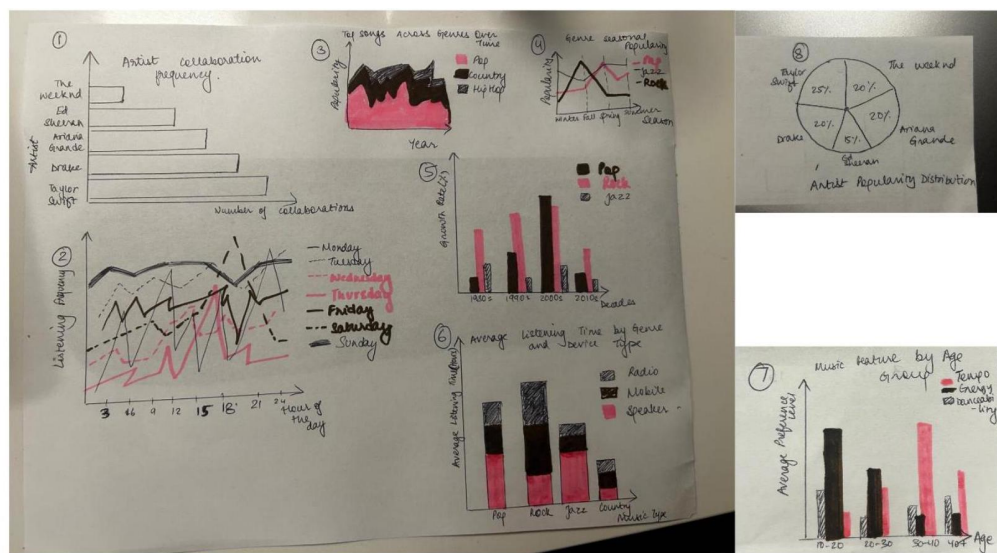


Figure 17: ..

Arrangement and Summary

- Begin with the Genre Popularity Heatmap to provide historical context on how genre popularity has changed over time.
- Follow with the Genre Evolution Tree to give a more detailed view of the development and intersection of genres over decades.
- Next, include the Bar Chart of Genre Preference by Age Group to introduce demographic influences on music tastes.
- Include the Map of the United States Based on Genre to show how music preferences vary regionally.

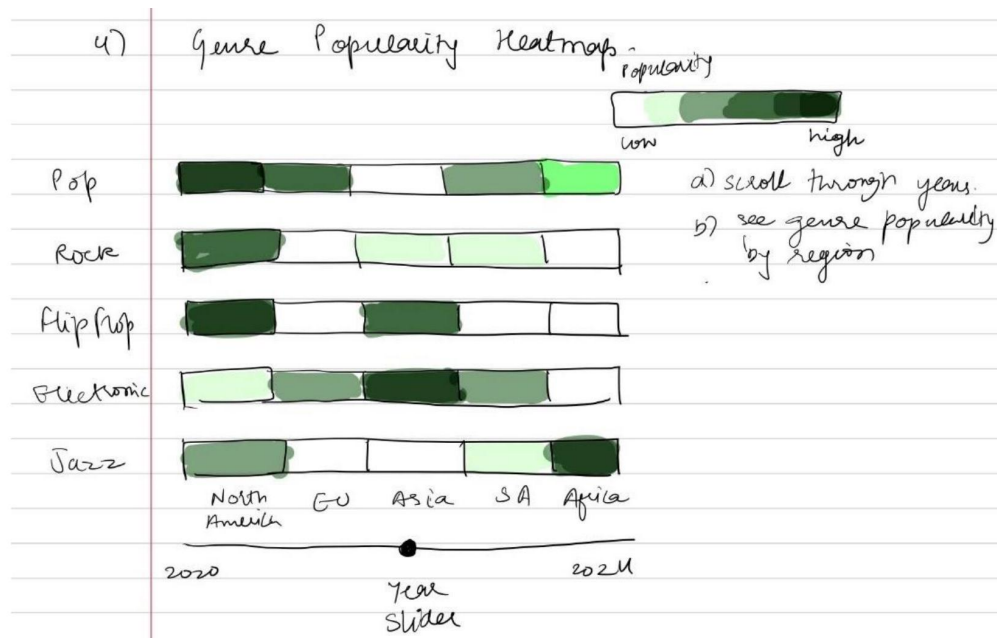


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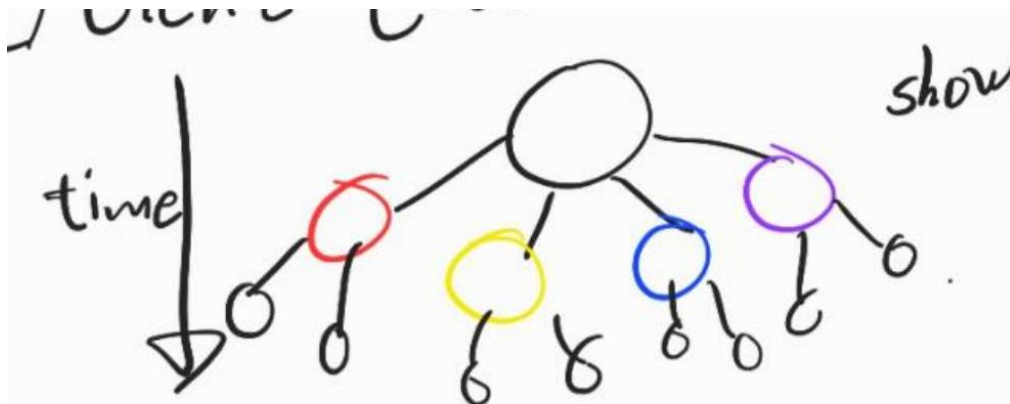


Figure 19: ..

- Show the Music Features by Age Group Chart to compare audio preferences, such as tempo and energy, across different listener demographics.
- End with the Lyrics Word Cloud to offer insights into common themes and language in popular songs.

The selected visualizations provide a structured and layered story of music trends. Starting with genre trends over decades, we set the historical stage, followed by demographic and regional insights that reveal deeper patterns in listener behavior. We conclude with an analysis of music features and lyrics, giving users a comprehensive view of evolving music tastes, preferences, and common themes in popular songs. This sequence captures the multifaceted landscape of music trends, artist influence, and listener behaviors.

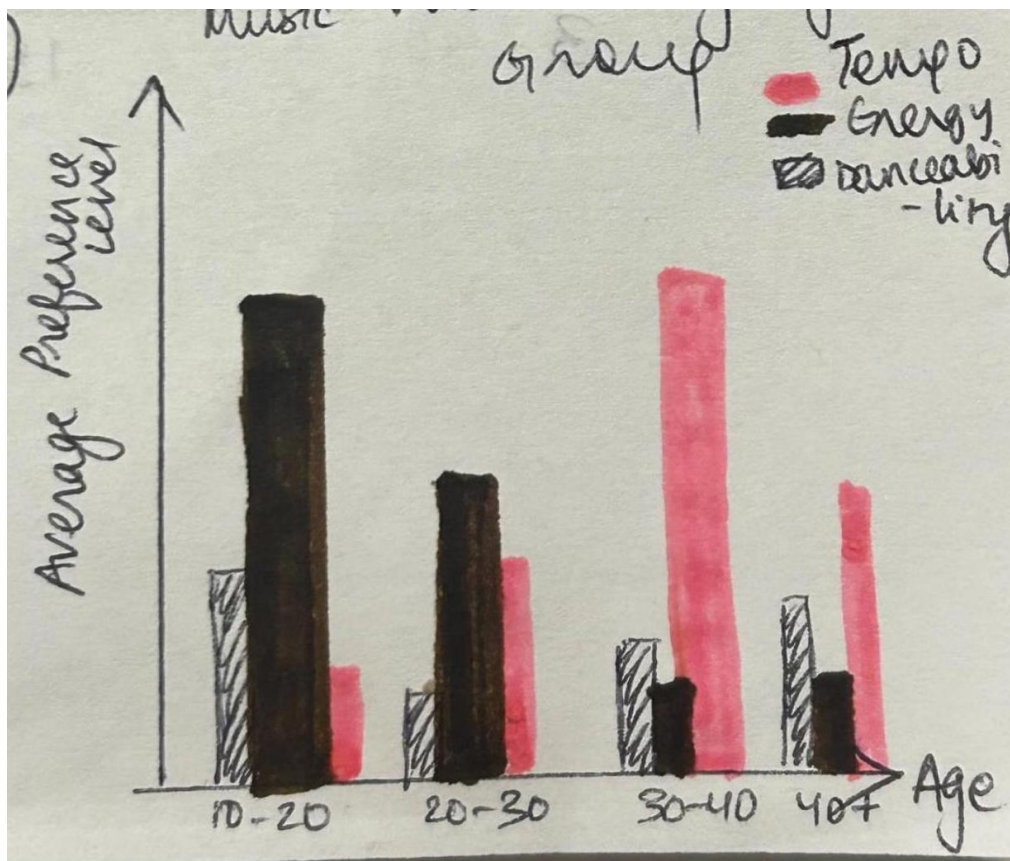


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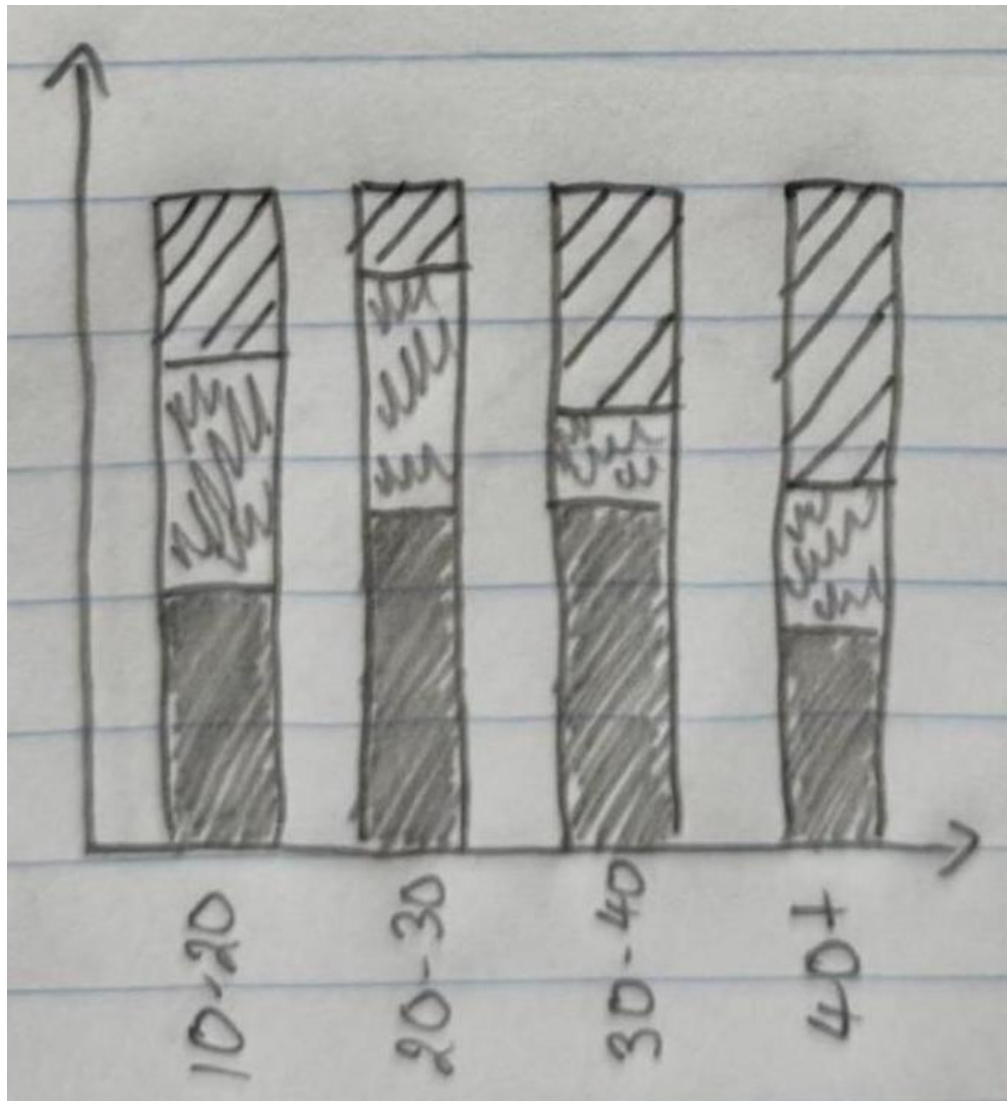


Figure 21: ..



Figure 22: ..

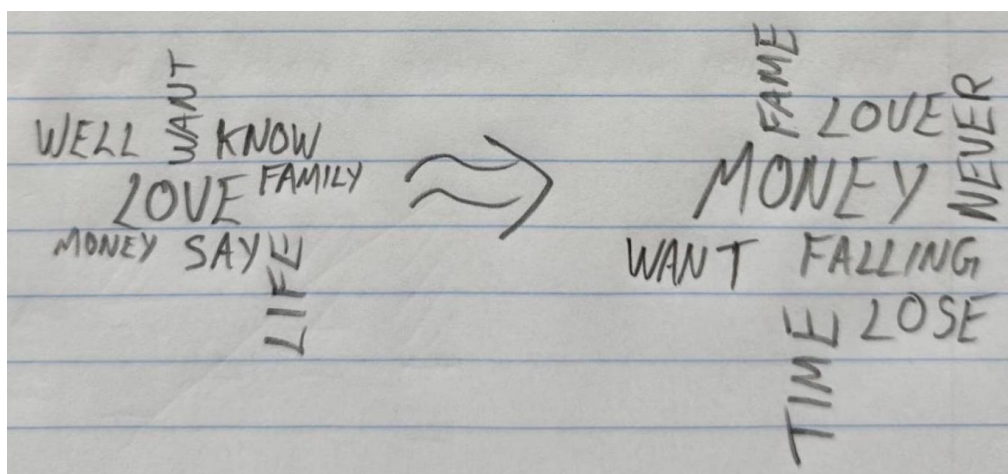


Figure 23: ..